

Satya

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Professional Summary

- AI/ML Engineer with 4 years of experience designing and deploying machine learning systems, generative AI pipelines, and scalable data engineering solutions across cloud-native environments.
- Hands-on expertise in LLM-powered applications, RAG architectures, multi-agent systems, and MLOps practices including model monitoring, CI/CD pipelines, and containerized deployments.
- Proven track record of building production-grade AI systems using Python, LangChain, LangGraph, OpenAI, AWS, and Azure with measurable improvements in model accuracy, latency, and operational efficiency.

Technical Skills

- **Programming Languages:** Python, SQL, Bash
- **Generative AI:** LangChain, LangGraph, LlamaIndex, RAG, Prompt Engineering, Multi-Agent Systems
- **LLM Providers:** OpenAI GPT-4, Claude, Gemini, Amazon Bedrock, Semantic Search
- **Machine Learning:** Scikit-learn, XGBoost, LightGBM, SHAP, Feature Engineering, Forecasting
- **Deep Learning:** TensorFlow, PyTorch, LSTM
- **MLOps & LLMOps:** MLflow, Azure ML, Model Monitoring, Drift Detection, A/B Testing, CI/CD
- **Data Engineering:** PySpark, Azure Data Factory, Azure Synapse Analytics, Azure DevOps, Snowflake
- **Cloud Platforms:** AWS (SageMaker, S3, Elastic Beanstalk), Microsoft Azure (ADF, Synapse, Azure ML)
- **Databases & Vector Stores:** PostgreSQL, FAISS, Pinecone, Delta Lake, Power BI
- **Infrastructure & APIs:** Docker, Kubernetes, FastAPI, Flask, REST APIs, GitHub Actions
- **Developer Tools:** Git, VS Code, Jupyter, Great Expectations, Postman

Work Experience

Intuit

Feb 2025 – Present

AI/ML Engineer

- Designed and deployed a multi-agent AI system using LangGraph and OpenAI GPT-4 to automate document understanding workflows, reducing manual review time by 35% across internal operations.
- Built a production-grade RAG pipeline with LlamaIndex, FAISS, and Amazon Bedrock, enabling semantic search over large enterprise knowledge bases with significantly improved retrieval relevance.
- Developed prompt engineering strategies and few-shot templates to improve LLM response quality, validated through automated evaluation frameworks and human review cycles.
- Orchestrated LangChain-based pipelines to connect retrieval, reasoning, and response generation components across multiple AI workflows in a unified agent architecture.
- Deployed AI microservices on AWS using FastAPI and Docker, handling 8K+ daily requests with average response latency reduced from 420ms to 180ms through async processing and caching.
- Implemented LLMops monitoring pipelines using MLflow to track hallucination rates, embedding drift, and throughput, enabling proactive detection of model degradation in production.
- Collaborated with platform engineering to containerize and orchestrate AI services on Kubernetes, supporting zero-downtime rolling deployments across multiple environments.
- Integrated Claude and Gemini APIs as fallback model layers within a model routing framework, reducing single-provider dependency and improving overall system resilience.
- Fine-tuned and served open-source LLMs on AWS SageMaker endpoints, reducing inference costs by 25% compared to third-party API usage while maintaining comparable response quality.
- Migrated vector storage from in-memory FAISS to Pinecone, enabling persistent and scalable similarity search across large document embedding collections.

Data & ML Engineer

- Built end-to-end ML pipelines for customer churn prediction using XGBoost and Scikit-learn, achieving 91% AUC-ROC and enabling proactive retention campaigns.
- Developed time-series forecasting models using LSTM and PyTorch to predict inventory demand, reducing forecast error by 18% compared to legacy heuristics.
- Trained LightGBM models for real-time risk scoring, performing feature engineering and A/B testing to validate model performance against production baselines.
- Engineered feature pipelines in PySpark on Azure ML to process large-scale transactional data, significantly reducing feature computation time for downstream model training.
- Deployed ML models as REST APIs using Flask and Docker on Azure, enabling real-time prediction serving integrated with existing business applications.
- Implemented MLflow experiment tracking and model registry to standardize model versioning, reproducibility, and handoff between data science and engineering teams.
- Automated data quality validation using Great Expectations and loaded processed datasets into Snowflake, improving reliability of data entering production ML pipelines.
- Built document classification models using TensorFlow, supporting multi-class labeling across a diverse set of document types.

Accenture**Dec 2020 – Jun 2022***Data Engineer*

- Designed and maintained scalable ETL pipelines using Azure Data Factory and PySpark to ingest and transform data from heterogeneous sources into Azure Synapse Analytics.
- Migrated on-premise data warehouse to Azure Synapse, reducing query execution time by 40% and enabling real-time analytics for downstream business intelligence teams.
- Developed automated data validation and reconciliation frameworks in Python and SQL to enforce data quality standards and improve pipeline reliability across ingestion layers.
- Implemented CI/CD pipelines for data workflows using Azure DevOps, cutting deployment cycles from 2 weeks to 3 days and improving overall release cadence.
- Built incremental data load patterns and partitioning strategies in Azure Synapse to optimize batch processing performance for large daily data volumes.
- Managed data storage and archival workflows using Delta Lake to support versioned, ACID-compliant data layers for downstream consumption.
- Collaborated with analytics teams to design star schema data models, enabling self-service reporting in Power BI and reducing ad-hoc data requests to the engineering team.

Projects

AI Knowledge Assistant | *Python, FastAPI, React, Gemini, FAISS, BM25, RAG*

- Built a production-style RAG application supporting PDF, text, and YouTube knowledge ingestion with hybrid semantic and keyword retrieval.
- Developed a FastAPI backend and React frontend using Gemini, FAISS, and BM25 to deliver grounded, source-backed responses with multi-user vector store isolation.

Customer Churn Prediction Platform | *Python, FastAPI, XGBoost, React, SHAP, Docker, AWS*

- Developed an end-to-end customer churn prediction platform using XGBoost, FastAPI, and React with SHAP-based model explainability and real-time prediction APIs.
- Implemented PostgreSQL prediction logging, statistical drift detection, Docker deployment on AWS Elastic Beanstalk, and automated CI/CD using GitHub Actions.

Education

University of Central Missouri

Master of Science in Information Technology

Jawaharlal Nehru Technological University

Bachelor of Technology in Electrical and Electronics Engineering

Certifications

- DeepLearning.AI – Natural Language Processing Specialization
- DeepLearning.AI – LangChain for LLM Application Development
- Microsoft Certified: Azure Data Engineer Associate